

An Evaluation of the Influence of Reproductive Factors on the Risk of Metastases from Uveal Melanoma

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Background: There is a paucity of data concerning the possible role played by hormonal factors in the risk of metastases from intraocular melanomas.

Methods: The authors studied the influence of post-diagnosis pregnancy and oral contraceptive use in a group of women of reproductive age (45 or younger) who were treated for uveal melanoma by proton beam irradiation. A baseline reproductive history had been collected before irradiation for all women, and interim reproductive data were collected by mailed questionnaire.

Results: In this age group, the overall rate of metastasis among women was similar to that of men treated during the same interval (adjusted rate ratio: 1.28; 95% confidence interval: 0.62–2.67). A total of 24 full-term pregnancies were reported among the 139 women still menstruating at diagnosis. Twenty-three women reported regular oral contraceptive use. Metastases developed in 15 of the 139 women. Compared with other women in the series, rates of metastases were not higher among the women who reported pregnancies ($P = 0.932$) or oral contraceptive use ($P = 0.424$) after diagnosis.

Conclusion: Although based on limited numbers, results suggest that the hormonal environment has no appreciable influence on risk of metastases in younger women with uveal melanoma. *Ophthalmology* 1993;100:1160–1166

Uveal melanoma is an uncommon tumor, particularly among younger persons. The disease is diagnosed in an estimated 150 women of reproductive age each year in the United States (Egan et al, unpublished data; presented at the ARVO Annual Meeting, Sarasota, May 1989). Low prevalence of disease has made it difficult to ascertain whether factors such as pregnancy and oral contraceptive use influence the risk of metastasis in these patients.

Pregnancy may theoretically influence melanoma-associated survival by one of several mechanisms. Pregnancy

is associated with a diminution of both cell-mediated and humoral immunity, which may allow proliferation of cancerous cells. Pregnancy also is accompanied by a general increase in circulating hormone levels, including estrogens, progesterone, and melanocyte-stimulating hormone. Melanocyte-stimulating hormone levels rise steadily throughout pregnancy with peak levels at term.¹ Activation of melanocytes by high circulating levels of melanocyte-stimulating hormone is thought to be responsible for the hyperpigmentation of the skin (chloasma) and benign changes of cutaneous nevi sometimes seen in pregnancy and during oral contraceptive use. Proliferative effects of these hormones either in pregnancy or with exogenous hormone use could worsen prognosis. Alternatively, prior pregnancy may impede metastases and improve survival by immunizing the patient against melanomas which may have antigens in common with fetal tissues.²

Numerous studies have examined the association of pregnancy with survival in cutaneous melanoma, as re-

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